

Assign 1 : write down 3 PEAS

Handwritten

1 - soldier *

Rest 2 from given topics.

PEAS : Performance Environment Actuator Sensor

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1. Playing a Tennis Match :

P → Points scored, Rally length, accuracy of shots, win/loss, stamina, court courage

E → tennis court, opponent behaviour, Court boundaries and net, ball.

A → Arm swings, leg movement, Grip, Jump, Communication with partner

S → Vision (Ball Trajectory), Hearing (referee calls), Perception.

2. Shopping for used AI books on the Internet

P → Best Price, Quality of Book listing, speed of search

E → Online market place, user reviews, seller ratings, Dynamic Prices

A → mouse clicks, typing queries, scrolling, adding to cart

S → screen display, text data, user ratings, images of books.

3. Soldier (Discussed in class)

P → mission success, enemy neutralization, survival, efficient use of resources

E → Battlefield, Terrain, weather conditions, obstacles

A → weapons (guns), movement (walk, run), communication devices

S → Cameras/vision, radar, GPS, microphones.

4. Playing Soccer

P → goals scored, passes, accuracy, stamina

E → soccer field, teammates, opponents, ball, weather

A → legs, arm, body movement, communication

S → eyes, ears, vision, touch, balance sensors.

5. Exploring the subsurface ocean of titanic

P → Depth Reached, area mapped, data accuracy, system safety

E → Subsurface ocean, darkness, chemical composition, temperature.

A → Thrusters, drills, Robotic arm, chemical analyzer

S → Sonar cameras, pressure sensors, temperature sensors.

6. Practicing Tennis against a wall

- P → accurate returns, consistent rally length, correct timing and speed
- E → Tennis court, wall, ball, gravity, lighting
- A → Arm and wrist movement, body movement
- S → vision, touch, body position

7. Performing a High Jump

- P → maximum high jump, skill improvement, safe landing
- E → Track field, bar, ground, weather conditions
- A → legs, arms, body muscles
- S → vision, balance sensors, muscle feedback

8. Knitting a Sweater

- P → correct pattern, No defects, Timely completion
- E → Yarn, Knitting needles, workspace
- A → Hands, fingers, Arm movements
- S → vision (stitches), touch (Yarn Tension)

9. Bidding on an item at an auction

- P → Win item at minimum cost, stay within budget, optimal bidding strategy
- E → Auction platform, other bidders, Time constraints
- A → Placing bids, adjusting bid amount
- S → Current highest bid, Time remaining, Competitor bidding behaviour